

TWILLINGATE, NL, Canada

WMO: 714020

Lat: 49.6880N

Lon: 54.8002W

Elev: 92

StdP: 100.22

Time Zone: -3.50 (NAF)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-15.2	-13.4	-19.6	0.7	-14.1	-18.0	0.8	-12.5	23.4	-4.3	20.2	-4.7	9.4	280	0.862

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.6	24.7	17.7	23.0	17.1	21.3	16.5	19.4	22.5	18.4	21.2	17.4	20.2	5.0	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.2	13.3	20.9	17.2	12.5	20.2	16.2	11.6	19.3	55.8	22.5	52.6	21.4	49.6	20.1	23.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.0	16.4	14.3	DB	-16.7	28.6	2.4	1.4	-18.4	29.6	-19.8	30.4	-21.1	31.1	-22.8	32.1	(4)
(5)			WB	-17.7	21.3	2.9	1.1	-19.8	22.1	-21.5	22.7	-23.2	23.3	-25.3	24.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.4	-5.1	-5.9	-3.5	0.9	5.1	10.1	15.8	16.0	12.1	6.9	2.3	-2.2	(6)	
	DBStd	8.42	3.96	4.40	3.81	3.27	4.07	4.75	3.93	3.19	3.43	3.29	3.38	3.67	(7)	
	HDD10.0	2563	467	445	417	274	163	58	6	1	16	107	231	378	(8)	
	HDD18.3	5102	725	678	676	524	411	249	97	86	187	353	480	637	(9)	
	CDD10.0	536	0	0	0	1	11	61	184	186	80	12	1	0	(10)	
	CDD18.3	32	0	0	0	0	0	2	17	12	2	0	0	0	(11)	
	CDH23.3	119	0	0	0	0	1	14	69	30	4	0	0	0	(12)	
	CDH26.7	10	0	0	0	0	0	2	7	1	0	0	0	0	(13)	
(14) Wind	WSAvg	7.0	8.6	8.1	7.6	6.9	6.1	5.6	5.2	5.4	6.6	7.3	7.8	8.4	(14)	
(15) Precipitation	PrecAvg	1419	140	126	116	107	97	97	102	98	122	139	134	141	(15)	
	PrecMax	1694	234	182	210	238	153	144	162	159	224	241	210	242	(16)	
	PrecMin	1125	79	66	51	45	41	47	64	47	53	42	60	62	(17)	
	PrecStd	138	40	29	34	44	27	25	26	29	40	45	36	44	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.7	8.1	7.7	14.4	20.5	25.2	27.8	26.1	23.5	18.1	14.2	9.7	(19)	
		MCWB	8.1	7.1	4.7	9.6	12.7	16.4	19.0	19.1	17.7	16.1	13.1	9.0	(20)	
	2%	DB	4.8	4.5	4.8	10.6	17.0	22.3	25.3	24.0	20.9	15.4	12.1	6.6	(21)	
		MCWB	4.0	3.8	2.7	7.4	10.9	15.3	17.8	17.8	16.9	13.0	11.0	6.0	(22)	
	5%	DB	2.1	1.8	3.0	7.9	14.3	19.9	23.4	22.4	19.0	13.5	9.7	4.3	(23)	
		MCWB	1.3	1.0	1.4	5.3	10.2	14.2	17.3	17.2	15.8	11.5	8.7	3.6	(24)	
	10%	DB	0.1	-0.3	1.3	5.6	11.8	17.4	21.5	20.9	17.3	11.8	7.2	2.6	(25)	
		MCWB	-0.5	-1.0	-0.2	3.4	8.5	13.2	16.8	16.6	14.4	9.9	6.0	1.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.1	7.4	6.0	11.0	14.6	18.9	21.0	20.6	19.1	16.6	13.3	9.0	(27)	
		MCDB	8.7	8.1	6.9	13.0	18.4	22.3	24.8	24.1	21.5	17.6	14.1	9.6	(28)	
	2%	WB	4.1	3.9	3.3	8.3	12.6	16.9	19.6	19.3	17.7	13.6	11.2	6.0	(29)	
		MCDB	4.7	4.5	4.1	10.4	15.3	20.2	23.0	22.2	19.9	14.9	11.9	6.6	(30)	
	5%	WB	1.4	1.1	1.5	5.7	10.6	15.2	18.6	18.3	16.3	11.8	8.8	3.7	(31)	
		MCDB	2.0	1.7	2.8	7.6	14.0	18.7	21.6	21.0	18.6	13.2	9.7	4.2	(32)	
	10%	WB	-0.5	-0.9	0.1	3.6	8.5	13.6	17.5	17.3	14.8	10.1	6.2	2.0	(33)	
		MCDB	0.0	-0.4	1.0	5.4	11.8	17.0	20.4	20.1	16.9	11.5	7.1	2.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.1	5.7	5.4	5.6	6.8	7.7	7.9	6.6	6.0	4.5	4.3	4.2	(35)	
		MCDBR	8.5	9.0	8.1	9.9	12.6	12.3	11.0	9.2	8.7	7.5	8.3	7.2	(36)	
	5% WB	MCWBR	8.8	8.9	7.0	7.3	7.7	6.6	5.5	4.9	5.7	6.3	8.2	7.6	(37)	
		MCWBR	8.5	9.1	7.6	9.5	11.8	11.1	9.6	8.5	8.3	7.4	8.4	7.1	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.293	0.290	0.314	0.348	0.363	0.379	0.403	0.380	0.356	0.333	0.307	0.302	(40)	
		taud	2.348	2.346	2.295	2.327	2.418	2.405	2.388	2.452	2.504	2.516	2.492	2.307	(41)	
	Edn at Noon		753	859	894	897	896	880	852	857	844	803	743	674	(42)	
		Edh at Noon	67	84	106	115	110	113	113	101	87	73	59	61	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.87	1.63	2.79	3.93	4.72	5.06	5.26	4.31	3.23	1.96	1.03	0.66	(44)	
	RadStd		0.07	0.22	0.32	0.45	0.42	0.56	0.43	0.30	0.25	0.22	0.11	0.05	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (10 neighbors)	N/A	N/A	+1.36	N/A	+0.45	+0.63	N/A	N/A	N/A	N/A

Nomenclature: See separate page